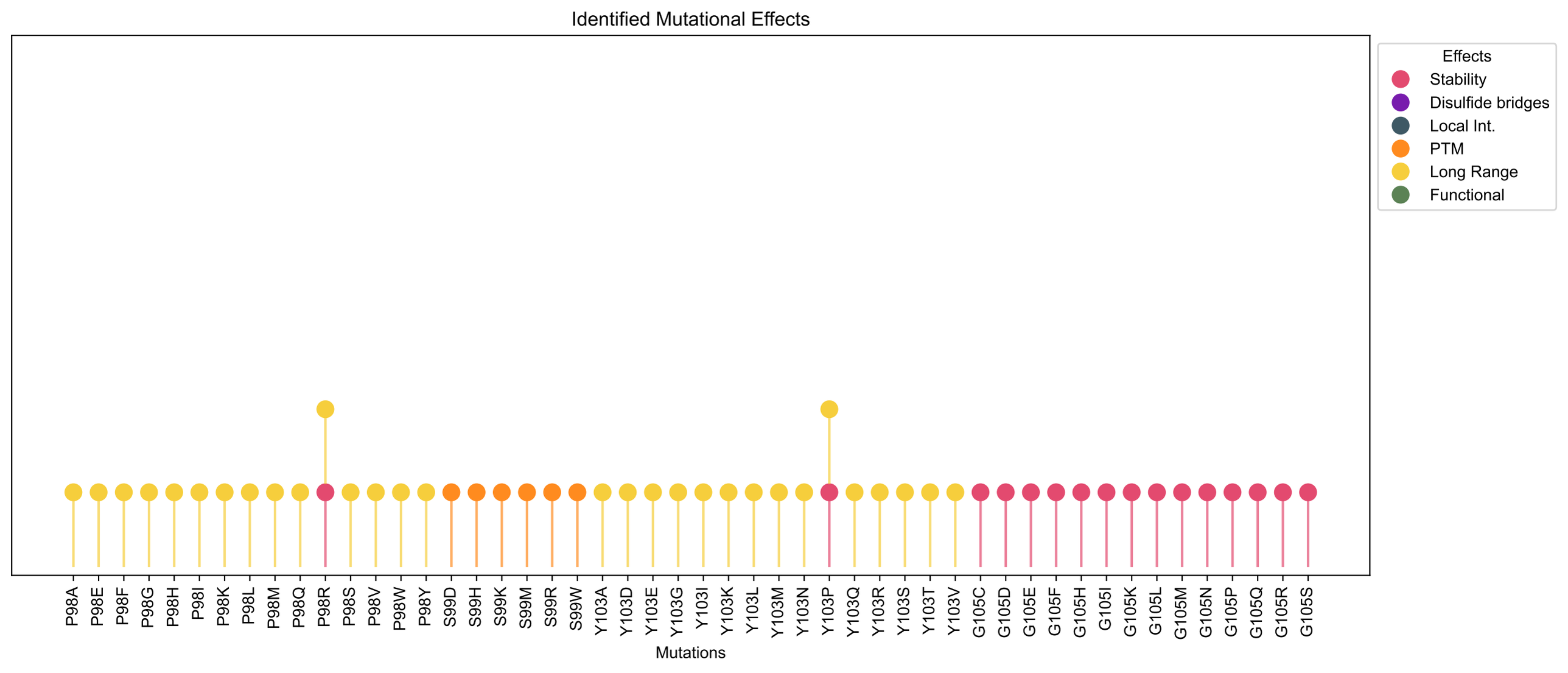
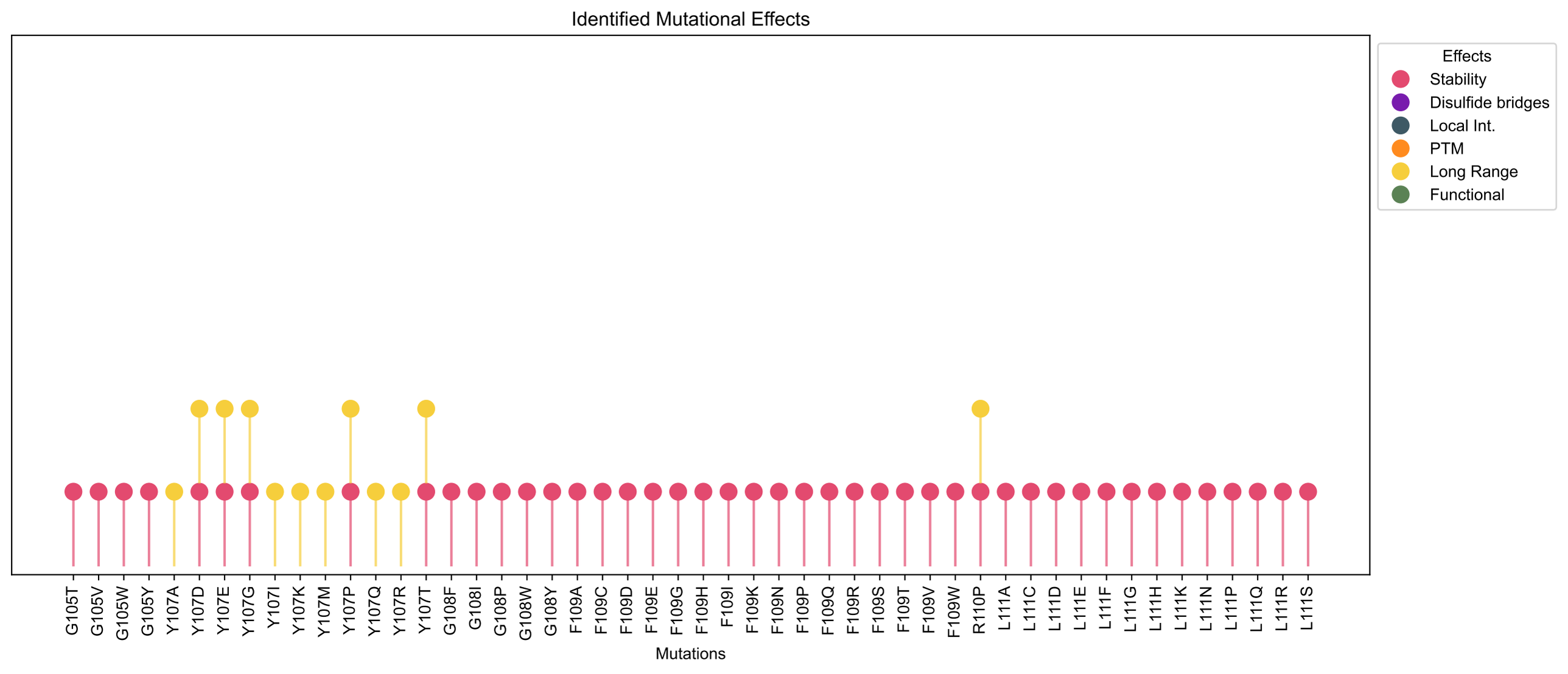
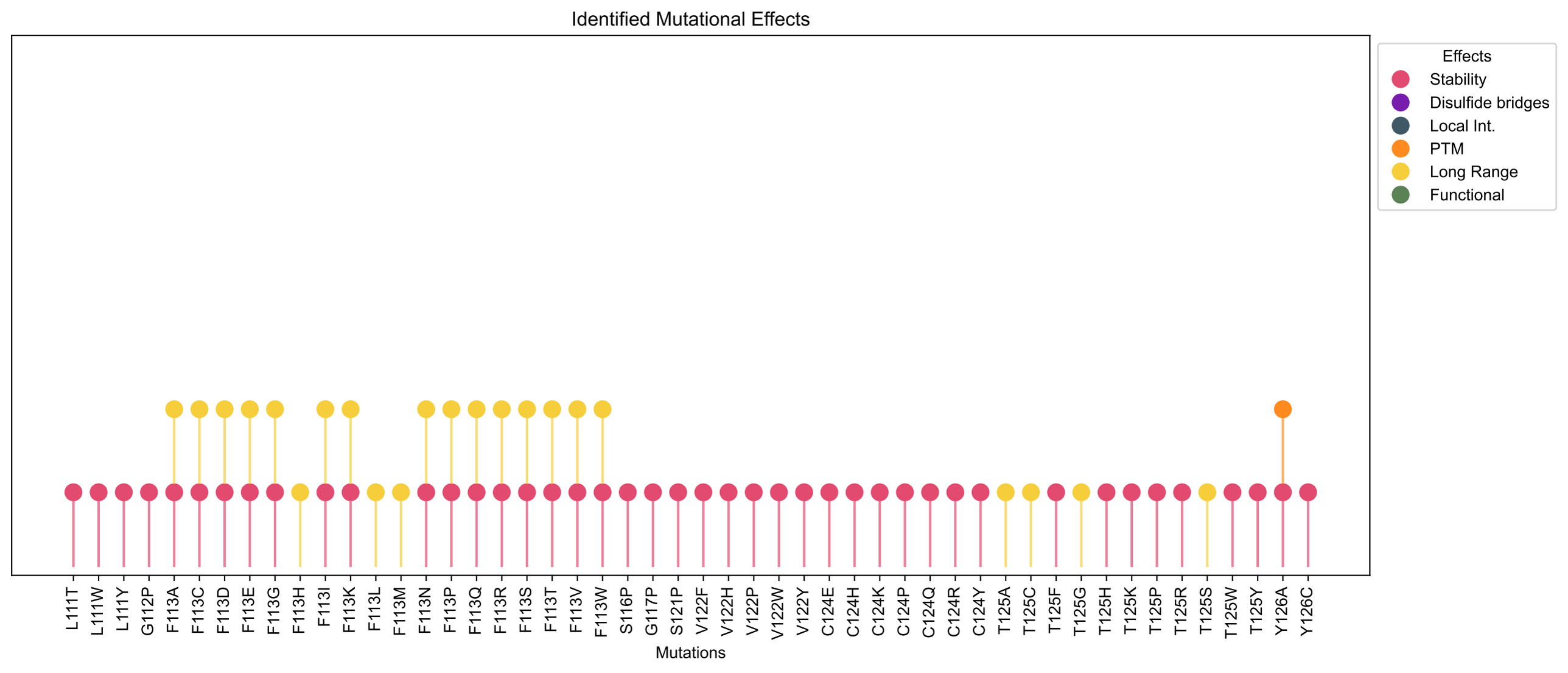
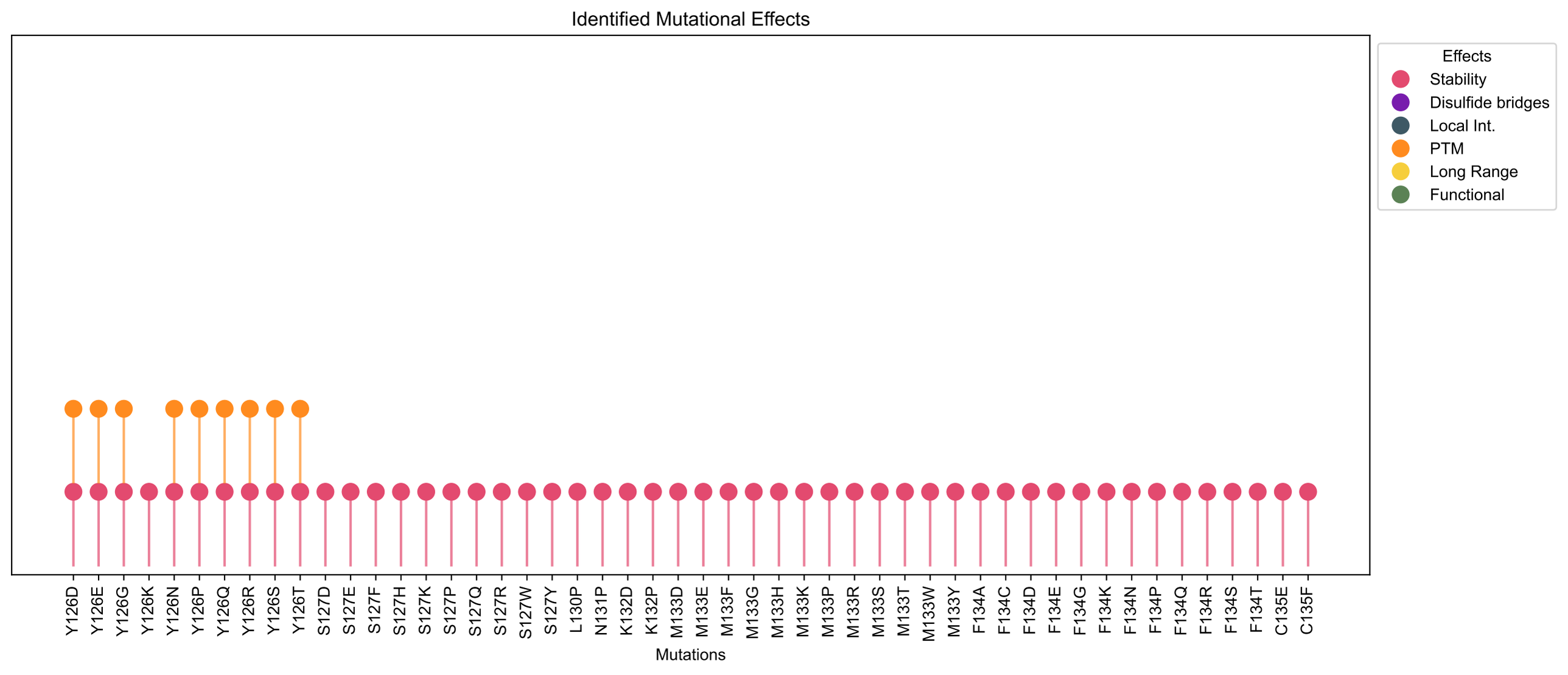
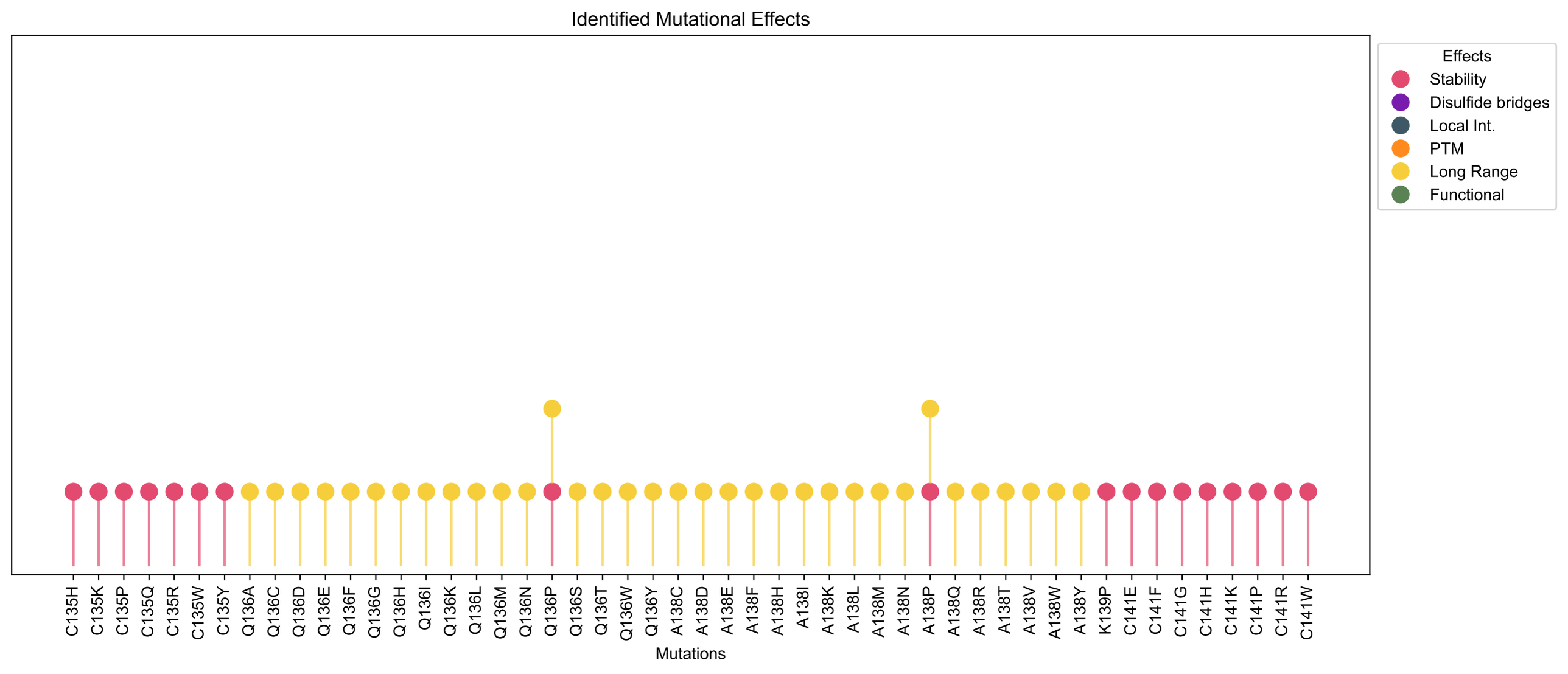
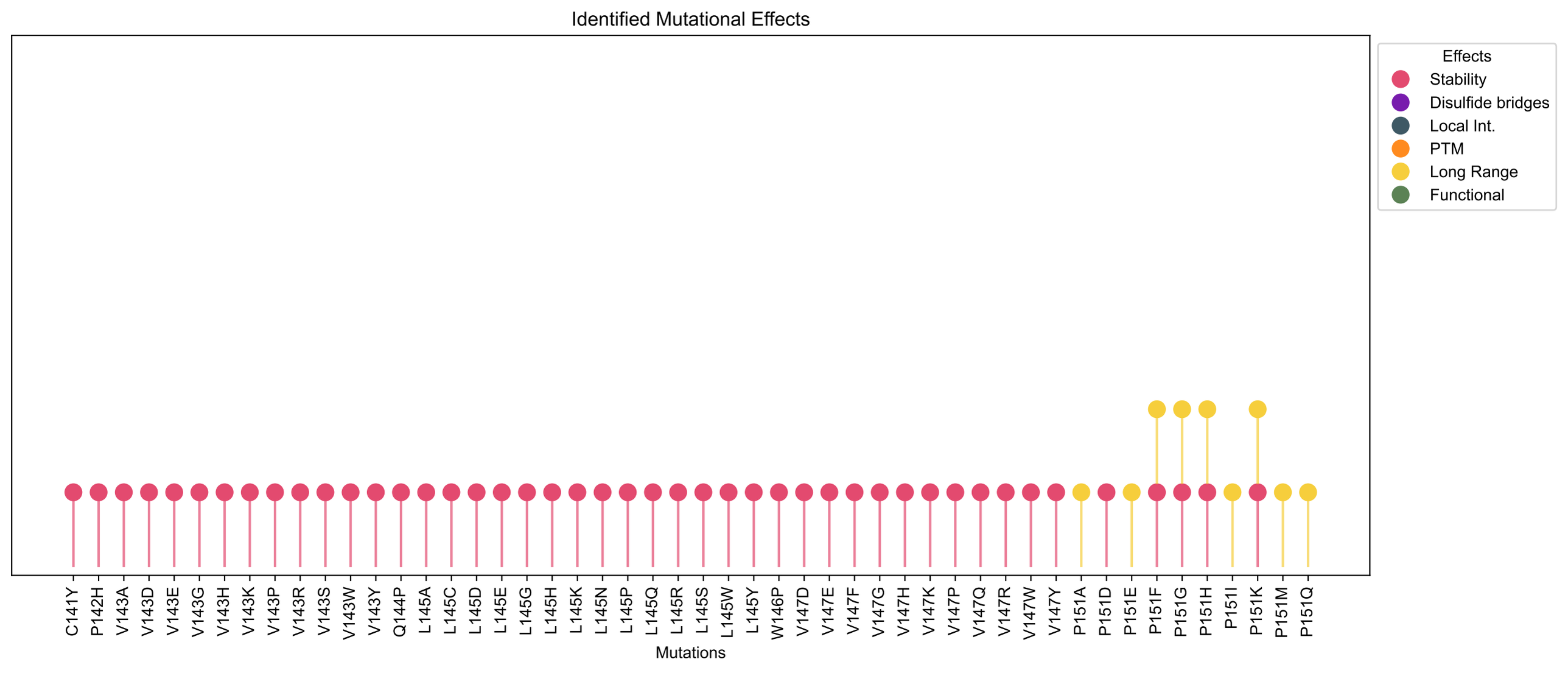

Identified Mutational Effects
Effects
Stability
Disulfide bridges
Local Int.
PTM
Long Range
Functional
P98A
P98E
P98F
P98G
P98H
P98I
P98K
P98L
P98M
P98Q
P98R
P98S
P98V
P98W
P98Y
S99D
S99H
S99K
S99M
S99R
S99W
Y103A
Y103D
Y103E
Y103G
Y103I
Y103K
Y103L
Y103M
Y103N
Y103P
Y103Q
Y103R
Y103S
Y103T
Y103V
G105C
G105D
G105E
G105F
G105H
G105I
G105K
G105L
G105M
G105N
G105P
G105Q
G105R
G105S
Mutations

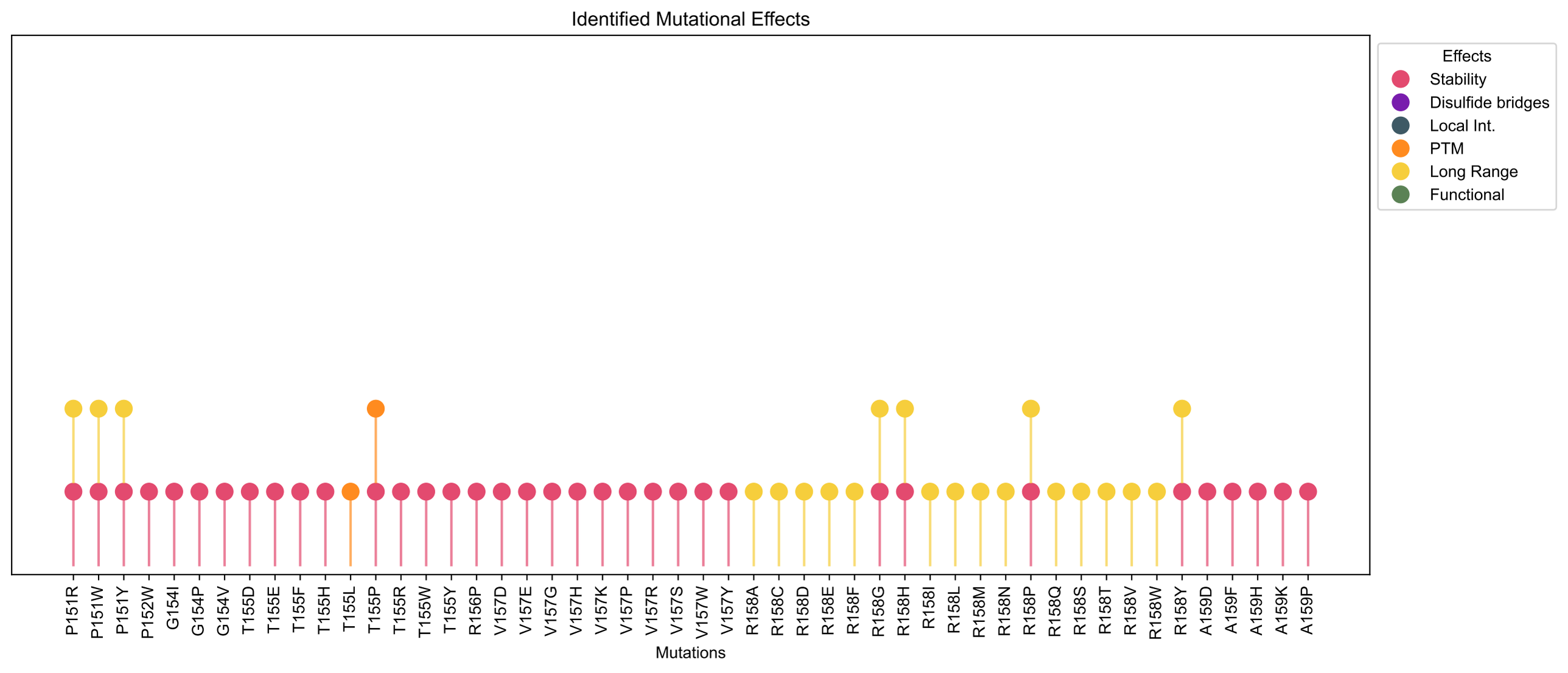

Identified Mutational Effects
Effects
Stability
Disulfide bridges
Local Int.
PTM
Long Range
Functional
G105T
G105V
G105W
G105Y
Y107A
Y107D
Y107E
Y107G
Y107I
Y107K
Y107M
Y107P
Y107Q
Y107R
Y107T
G108F
G108I
G108P
G108W
G108Y
F109A
F109C
F109D
F109E
F109G
F109H
F109I
F109K
F109N
F109P
F109Q
F109R
F109S
F109T
F109V
F109W
R110P
L111A
L111C
L111D
L111E
L111F
L111G
L111H
L111K
L111N
L111P
L111Q
L111R
L111S
Mutations

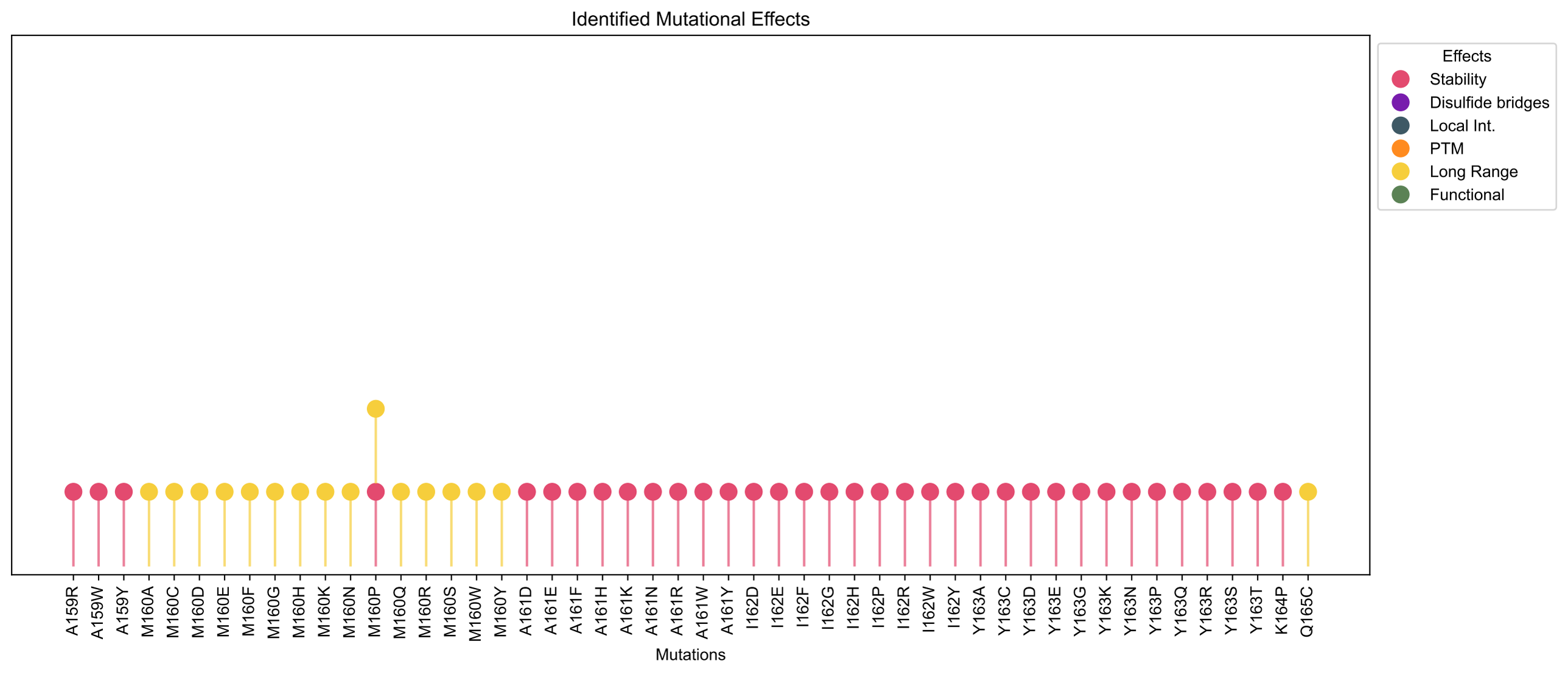

Identified Mutational Effects
Effects
Stability
Disulfide bridges
Local Int.
PTM
Long Range
Functional
L111T
L111W
L111Y
G112P
F113A
F113C
F113D
F113E
F113G
F113H
F113I
F113K
F113L
F113M
F113N
F113P
F113Q
F113R
F113S
F113T
F113V
F113W
S116P
G117P
S121P
V122F
V122H
V122P
V122W
V122Y
C124E
C124H
C124K
C124P
C124Q
C124R
C124Y
T125A
T125C
T125F
T125G
T125H
T125K
T125P
T125R
T125S
T125W
T125Y
Y126A
Y126C
Mutations

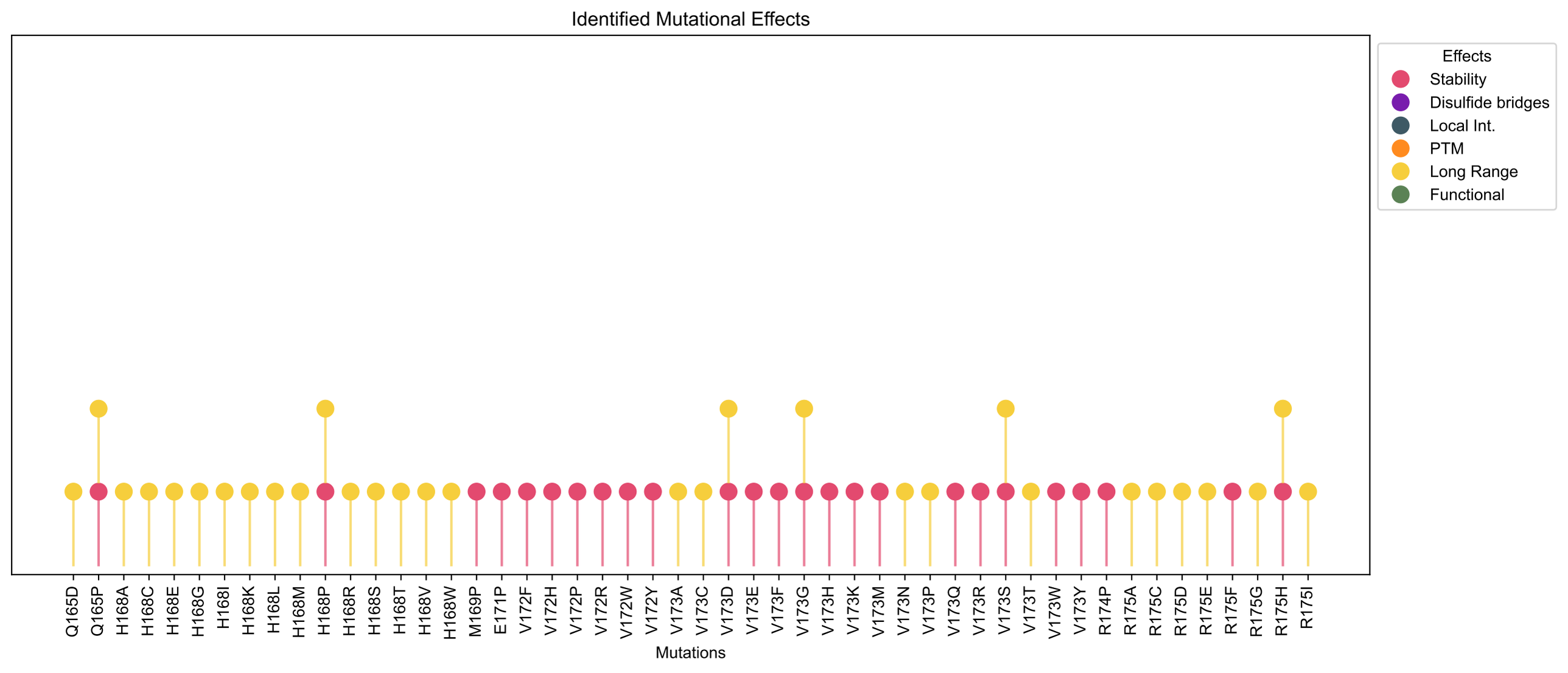

Identified Mutational Effects
Effects
Stability
Disulfide bridges
Local Int.
PTM
Long Range
Functional
Y126D
Y126E
Y126G
Y126K
Y126N
Y126P
Y126Q
Y126R
Y126S
Y126T
S127D
S127E
S127F
S127H
S127K
S127P
S127Q
S127R
S127W
S127Y
L130P
N131P
K132D
K132P
M133D
M133E
M133F
M133G
M133H
M133K
M133P
M133R
M133S
M133T
M133W
M133Y
F134A
F134C
F134D
F134E
F134G
F134K
F134N
F134P
F134Q
F134R
F134S
F134T
C135E
C135F
Mutations

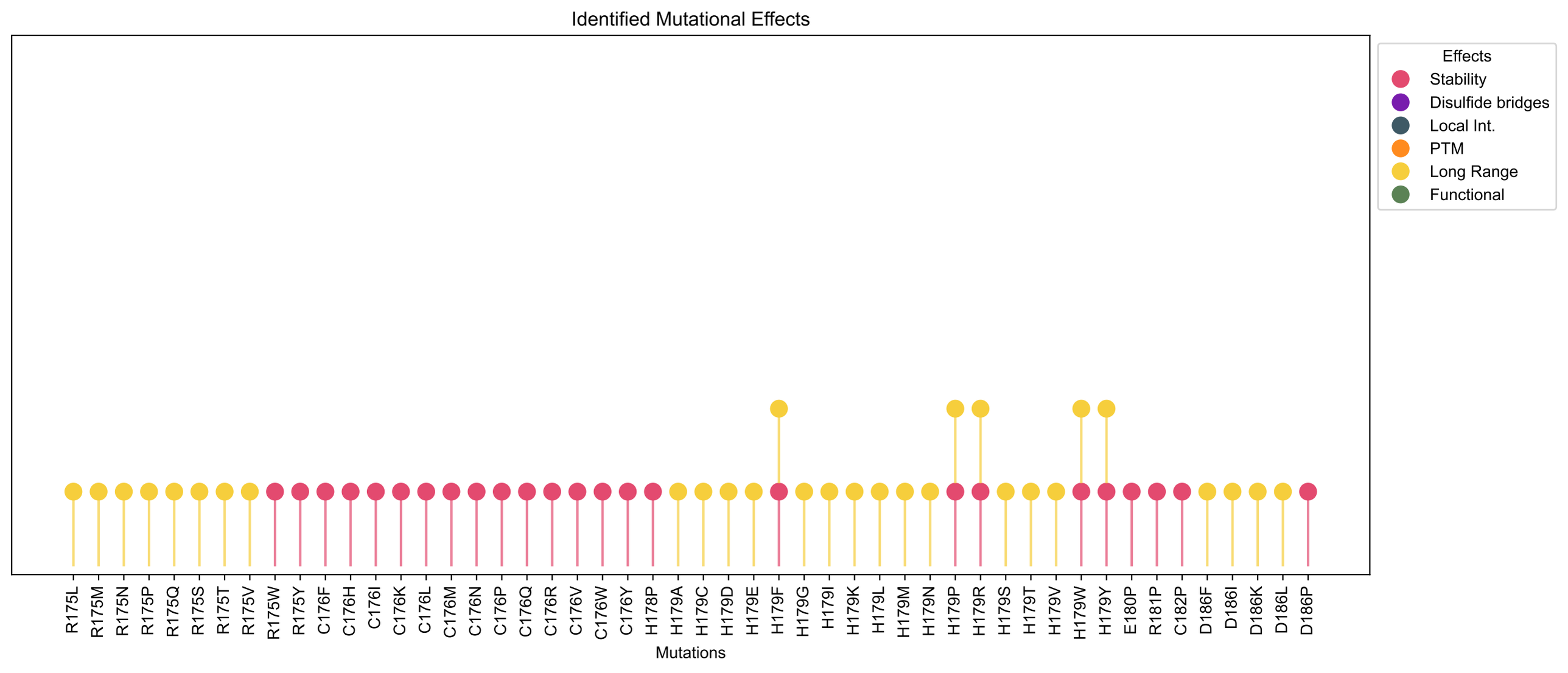

Identified Mutational Effects
Effects
Stability
Disulfide bridges
Local Int.
PTM
Long Range
Functional
C135H
C135K
C135P
C135Q
C135R
C135W
C135Y
Q136A
Q136C
Q136D
Q136E
Q136F
Q136G
Q136H
Q136I
Q136K
Q136L
Q136M
Q136N
Q136P
Q136S
Q136T
Q136W
Q136Y
A138C
A138D
A138E
A138F
A138H
A138I
A138K
A138L
A138M
A138N
A138P
A138Q
A138R
A138T
A138V
A138W
A138Y
K139P
C141E
C141F
C141G
C141H
C141K
C141P
C141R
C141W
Mutations

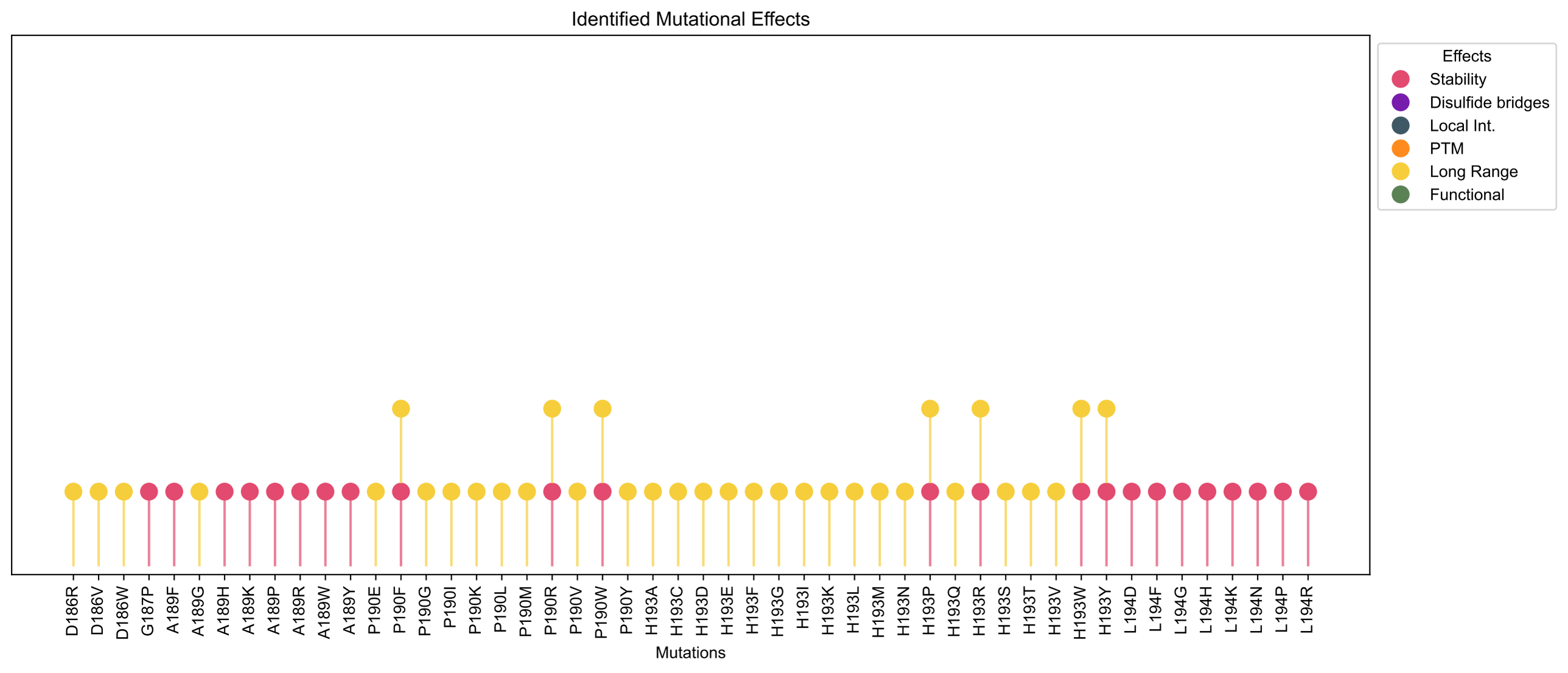

Identified Mutational Effects
Effects
Stability
Disulfide bridges
Local Int.
PTM
Long Range
Functional
C141Y
P142H
V143A
V143D
V143E
V143G
V143H
V143K
V143P
V143R
V143S
V143W
V143Y
Q144P
L145A
L145C
L145D
L145E
L145G
L145H
L145K
L145N
L145P
L145Q
L145R
L145S
L145W
L145Y
W146P
V147D
V147E
V147F
V147G
V147H
V147K
V147P
V147Q
V147R
V147W
V147Y
P151A
P151D
P151E
P151F
P151G
P151H
P151I
P151K
P151M
P151Q
Mutations

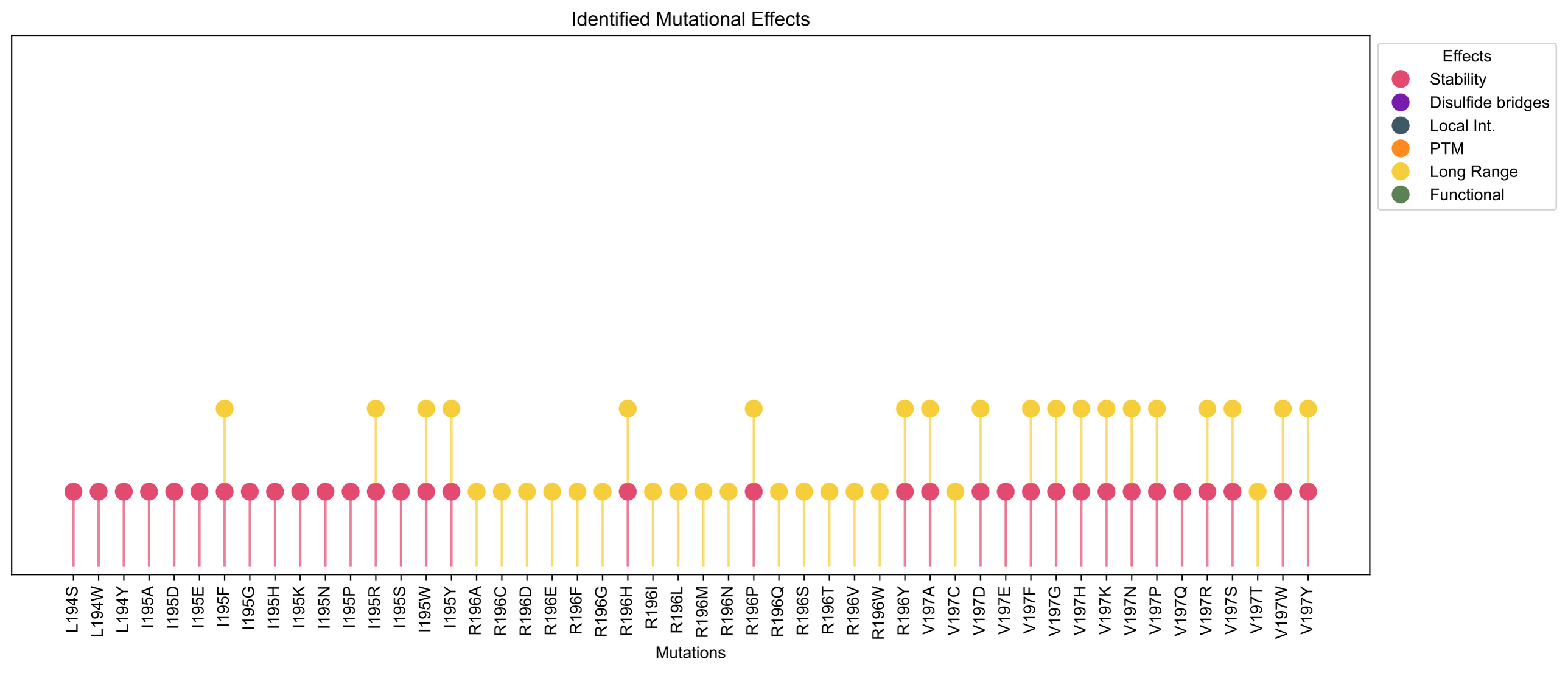

Identified Mutational Effects
Effects
Stability
Disulfide bridges
Local Int.
PTM
Long Range
Functional
P151R
P151W
P151Y
P152W
G154I
G154P
G154V
T155D
T155E
T155F
T155H
T155L
T155P
T155R
T155W
T155Y
R156P
V157D
V157E
V157G
V157H
V157K
V157P
V157R
V157S
V157W
V157Y
R158A
R158C
R158D
R158E
R158F
R158G
R158H
R158I
R158L
R158M
R158N
R158P
R158Q
R158S
R158T
R158V
R158W
R158Y
A159D
A159F
A159H
A159K
A159P
Mutations

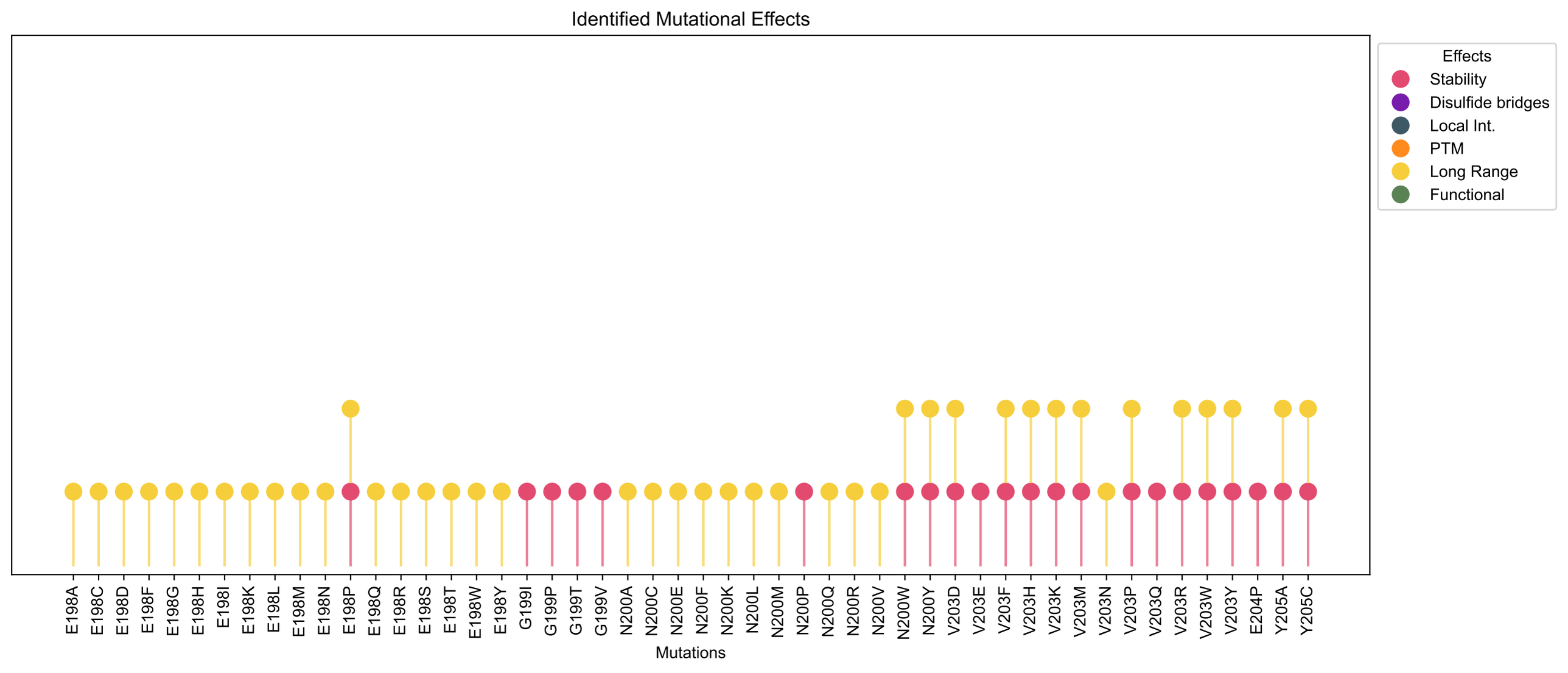

Identified Mutational Effects
Effects
Stability
Disulfide bridges
Local Int.
PTM
Long Range
Functional
A159R
A159W
A159Y
M160A
M160C
M160D
M160E
M160F
M160G
M160H
M160K
M160N
M160P
M160Q
M160R
M160S
M160W
M160Y
A161D
A161E
A161F
A161H
A161K
A161N
A161R
A161W
A161Y
I162D
I162E
I162F
I162G
I162H
I162P
I162R
I162W
I162Y
Y163A
Y163C
Y163D
Y163E
Y163G
Y163K
Y163N
Y163P
Y163Q
Y163R
Y163S
Y163T
K164P
Q165C
Mutations

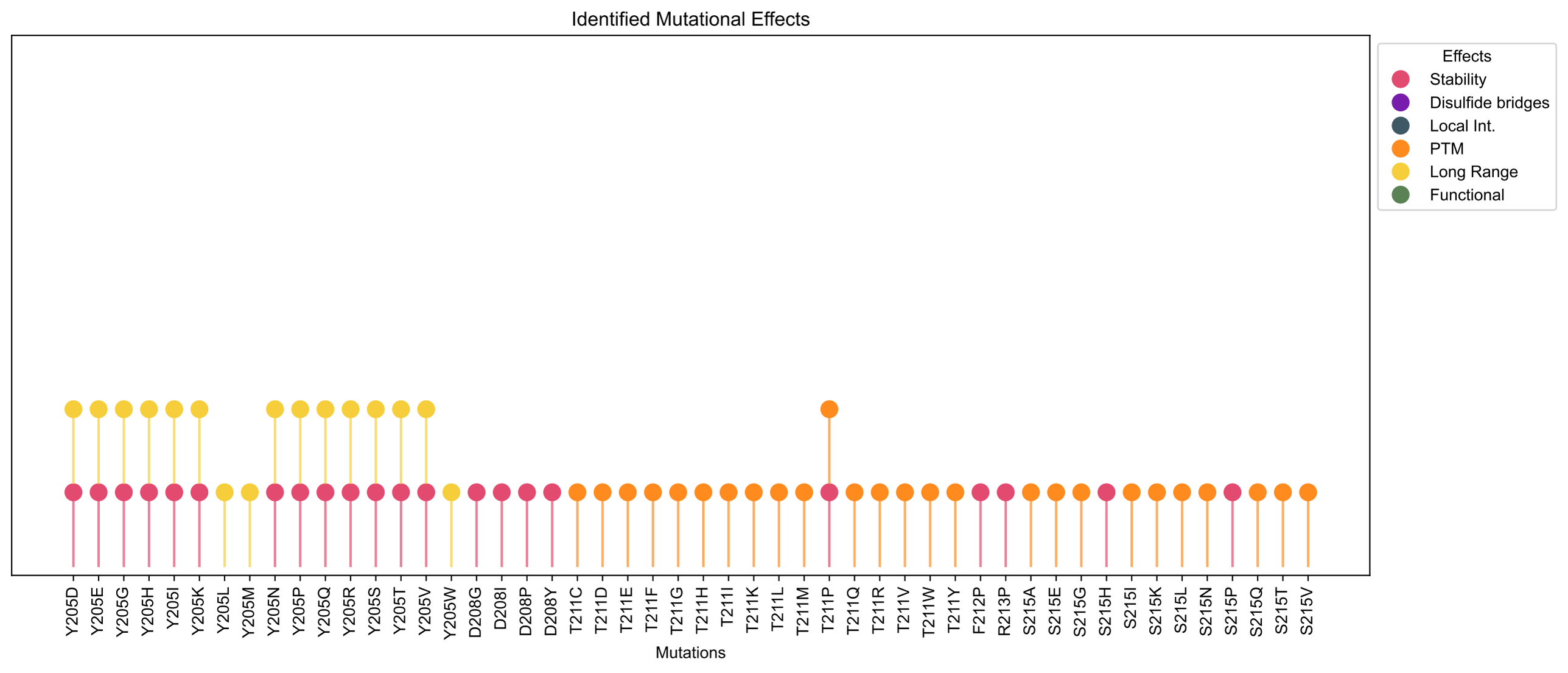

Identified Mutational Effects
Effects
Stability
Disulfide bridges
Local Int.
PTM
Long Range
Functional
Q165D
Q165P
H168A
H168C
H168E
H168G
H168I
H168K
H168L
H168M
H168P
H168R
H168S
H168T
H168V
H168W
M169P
E171P
V172F
V172H
V172P
V172R
V172W
V172Y
V173A
V173C
V173D
V173E
V173F
V173G
V173H
V173K
V173M
V173N
V173P
V173Q
V173R
V173S
V173T
V173W
V173Y
R174P
R175A
R175C
R175D
R175E
R175F
R175G
R175H
R175I
Mutations

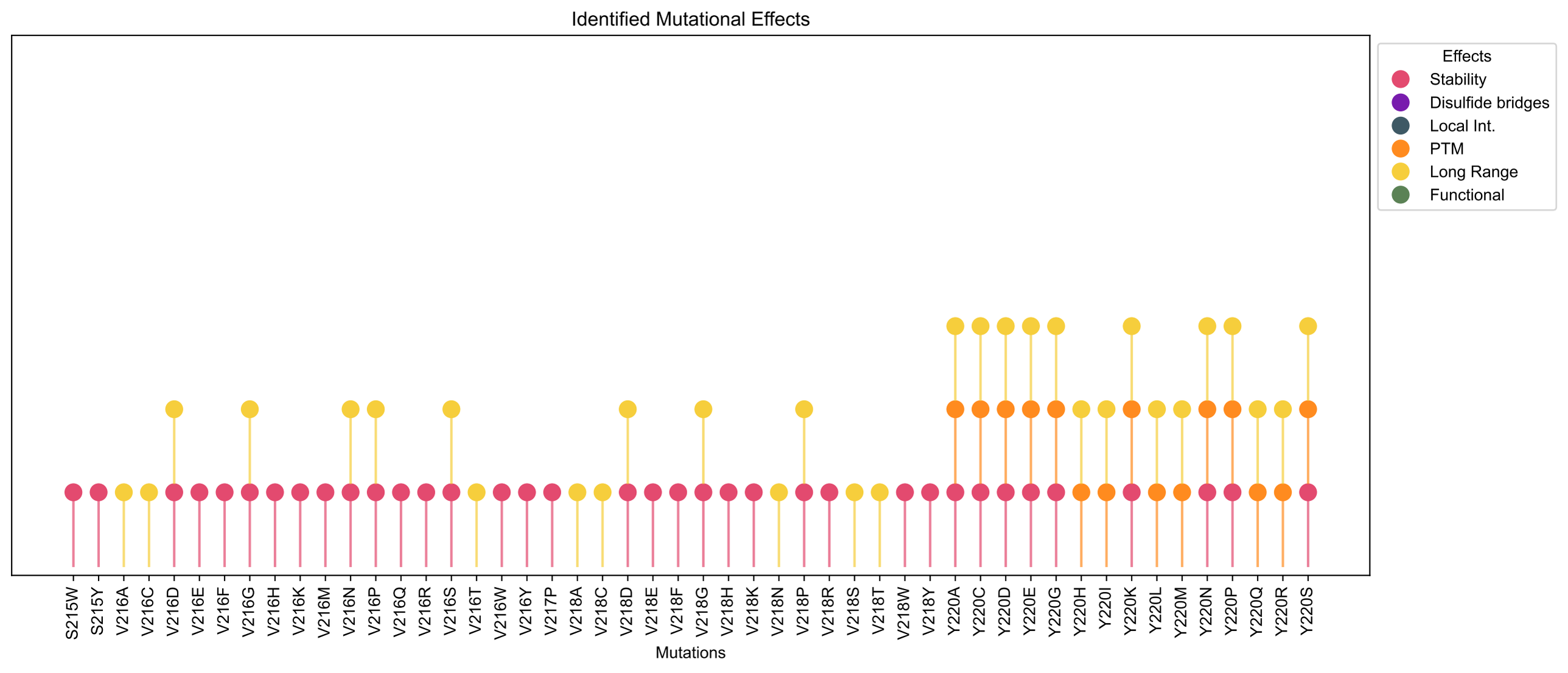

Identified Mutational Effects
Effects
Stability
Disulfide bridges
Local Int.
PTM
Long Range
Functional
R175L
R175M
R175N
R175P
R175Q
R175S
R175T
R175V
R175W
R175Y
C176F
C176H
C176I
C176K
C176L
C176M
C176N
C176P
C176Q
C176R
C176V
C176W
C176Y
H178P
H179A
H179C
H179D
H179E
H179F
H179G
H179I
H179K
H179L
H179M
H179N
H179P
H179R
H179S
H179T
H179V
H179W
H179Y
E180P
R181P
C182P
D186F
D186I
D186K
D186L
D186P
Mutations

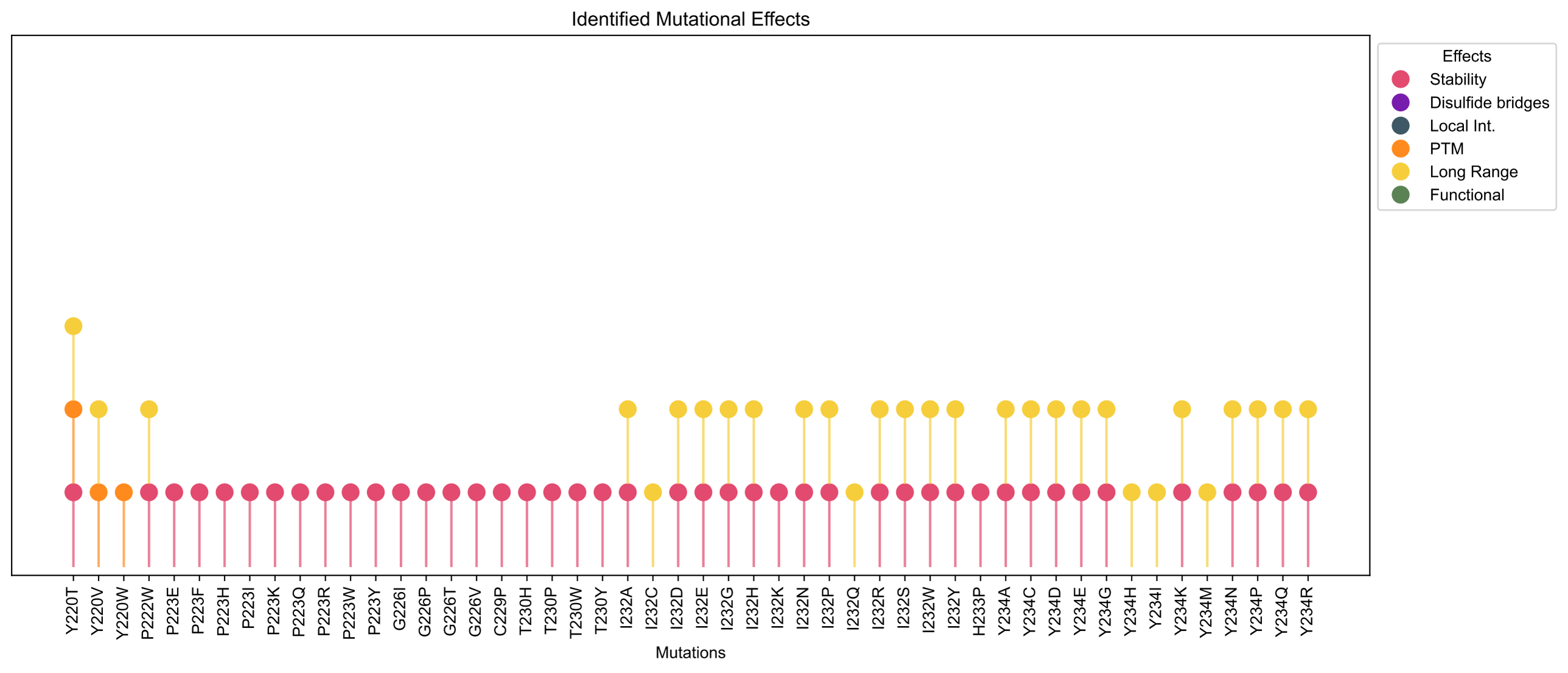

Identified Mutational Effects
Effects
Stability
Disulfide bridges
Local Int.
PTM
Long Range
Functional
D186R
D186V
D186W
G187P
A189F
A189G
A189H
A189K
A189P
A189R
A189W
A189Y
P190E
P190F
P190G
P190I
P190K
P190L
P190M
P190R
P190V
P190W
P190Y
H193A
H193C
H193D
H193E
H193F
H193G
H193I
H193K
H193L
H193M
H193N
H193P
H193Q
H193R
H193S
H193T
H193V
H193W
H193Y
L194D
L194F
L194G
L194H
L194K
L194N
L194P
L194R
Mutations

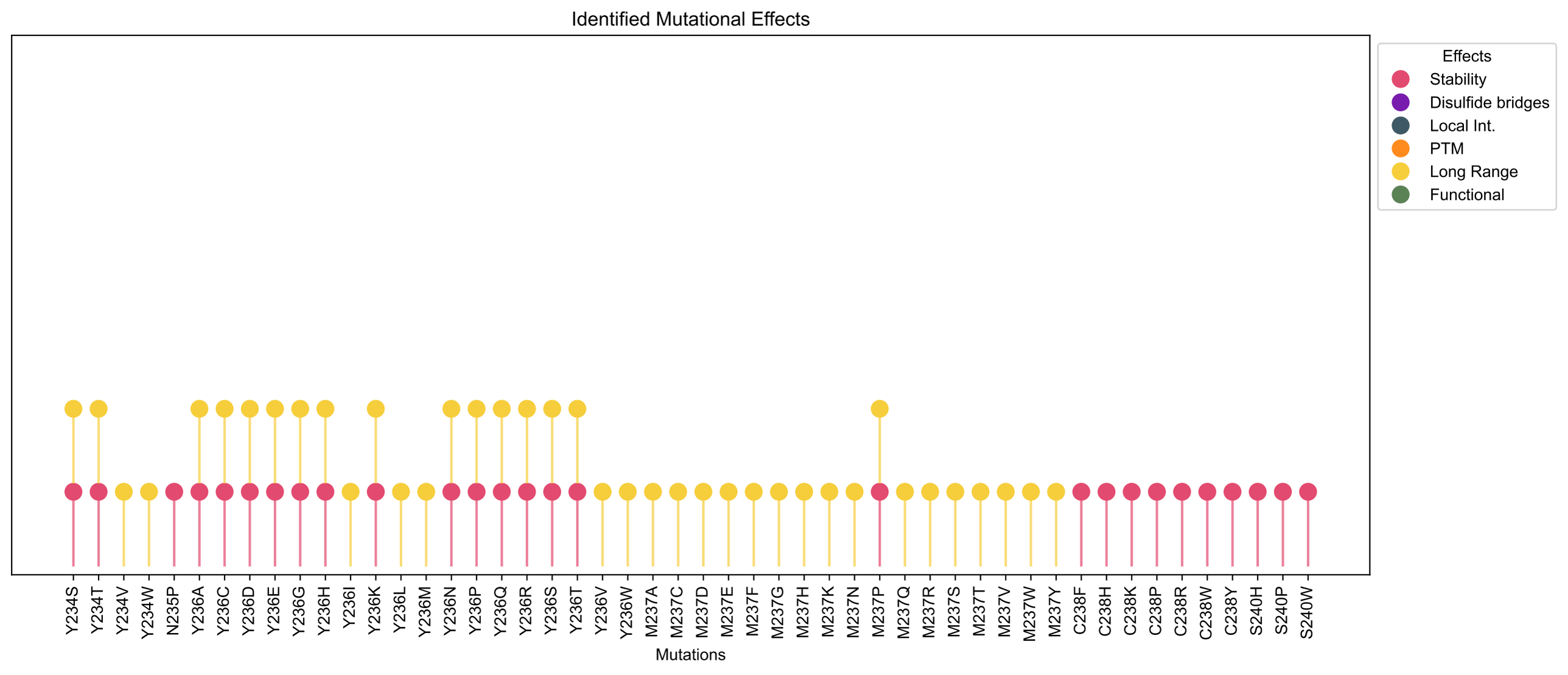

Identified Mutational Effects
Effects
Stability
Disulfide bridges
Local Int.
PTM
Long Range
Functional
L194S
L194W
L194Y
I195A
I195D
I195E
I195F
I195G
I195H
I195K
I195N
I195P
I195R
I195S
I195W
I195Y
R196A
R196C
R196D
R196E
R196F
R196G
R196H
R196I
R196L
R196M
R196N
R196P
R196Q
R196S
R196T
R196V
R196W
R196Y
V197A
V197C
V197D
V197E
V197F
V197G
V197H
V197K
V197N
V197P
V197Q
V197R
V197S
V197T
V197W
V197Y
Mutations

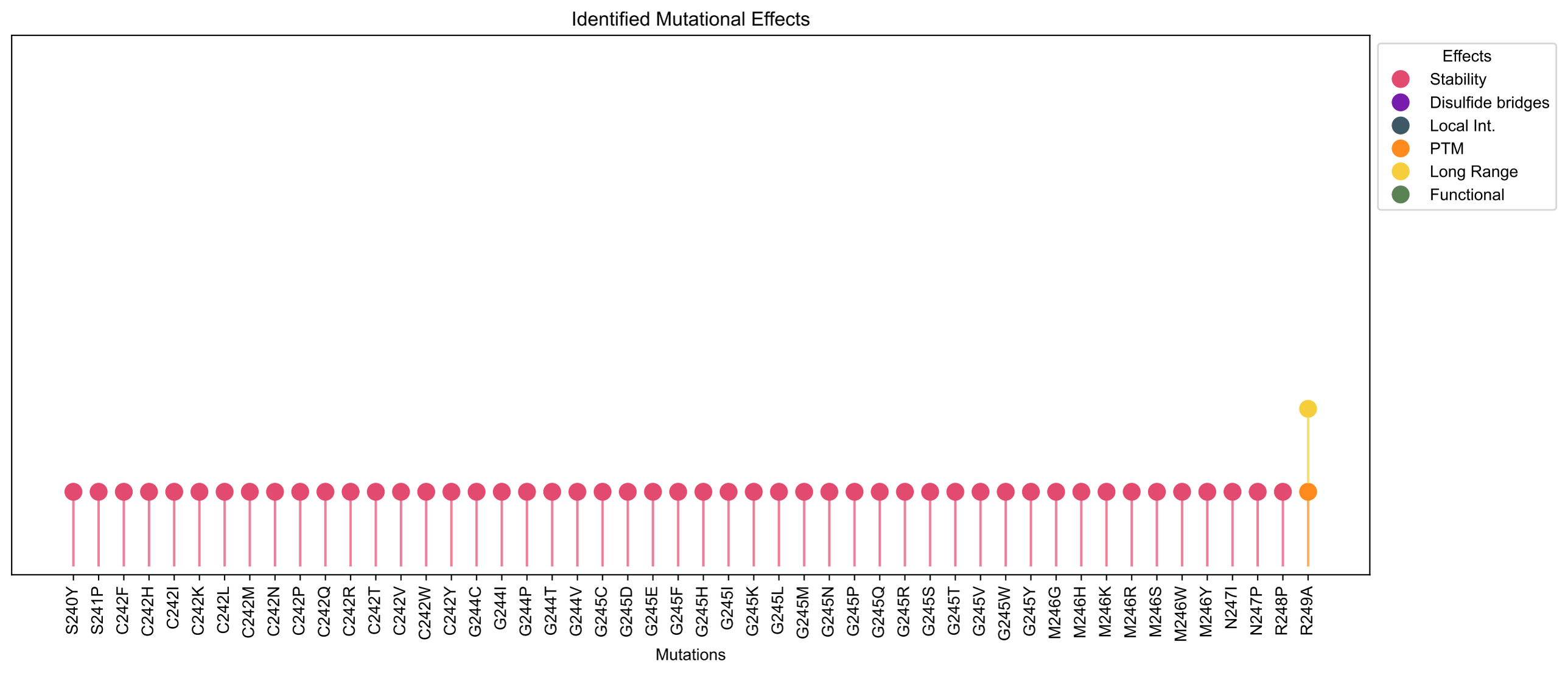

Identified Mutational Effects
Effects
Stability
Disulfide bridges
Local Int.
PTM
Long Range
Functional
E198A
E198C
E198D
E198F
E198G
E198H
E198I
E198K
E198L
E198M
E198N
E198P
E198Q
E198R
E198S
E198T
E198W
E198Y
G199I
G199P
G199T
G199V
N200A
N200C
N200E
N200F
N200K
N200L
N200M
N200P
N200Q
N200R
N200V
N200W
N200Y
V203D
V203E
V203F
V203H
V203K
V203M
V203N
V203P
V203Q
V203R
V203W
V203Y
E204P
Y205A
Y205C
Mutations

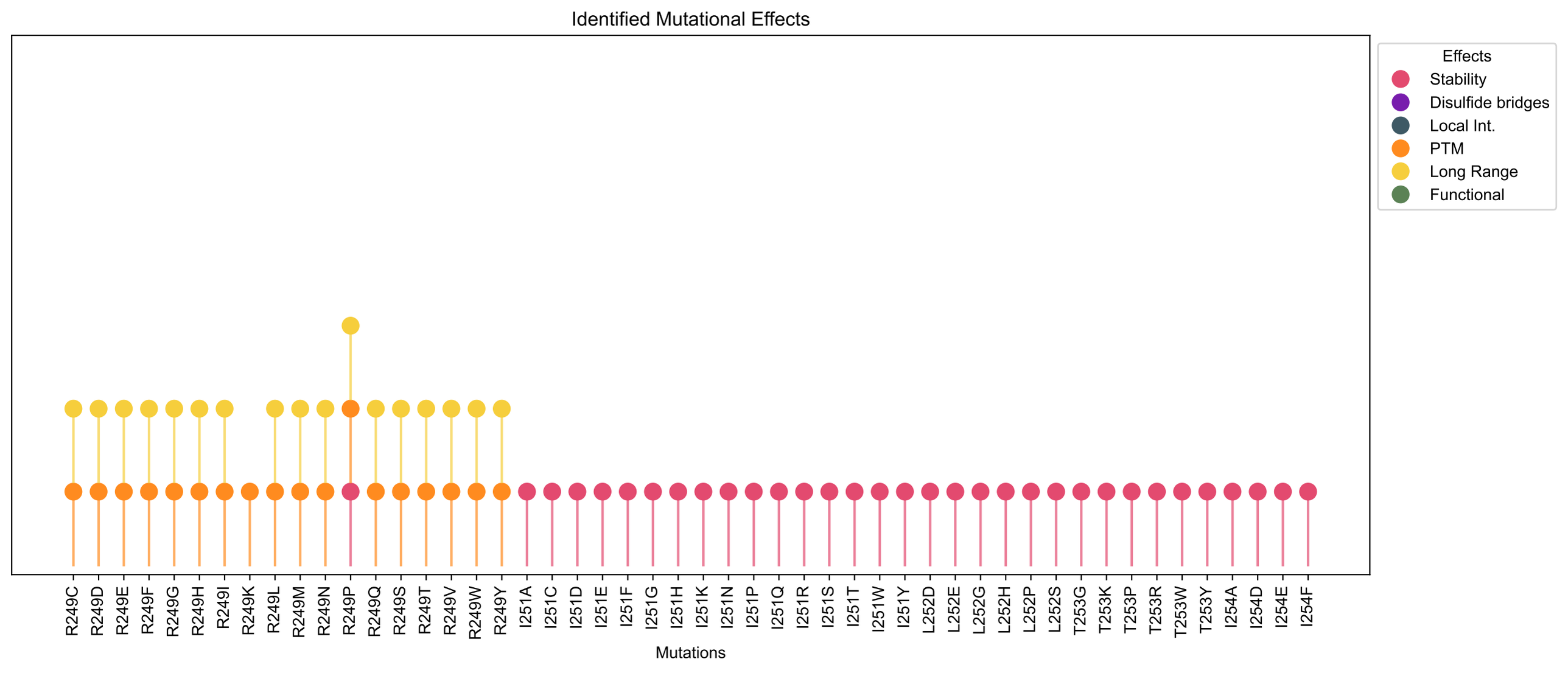

Identified Mutational Effects
Effects
Stability
Disulfide bridges
Local Int.
PTM
Long Range
Functional
Y205D
Y205E
Y205G
Y205H
Y205I
Y205K
Y205L
Y205M
Y205N
Y205P
Y205Q
Y205R
Y205S
Y205T
Y205V
Y205W
D208G
D208I
D208P
D208Y
T211C
T211D
T211E
T211F
T211G
T211H
T211I
T211K
T211L
T211M
T211P
T211Q
T211R
T211V
T211W
T211Y
F212P
R213P
S215A
S215E
S215G
S215H
S215I
S215K
S215L
S215N
S215P
S215Q
S215T
S215V
Mutations

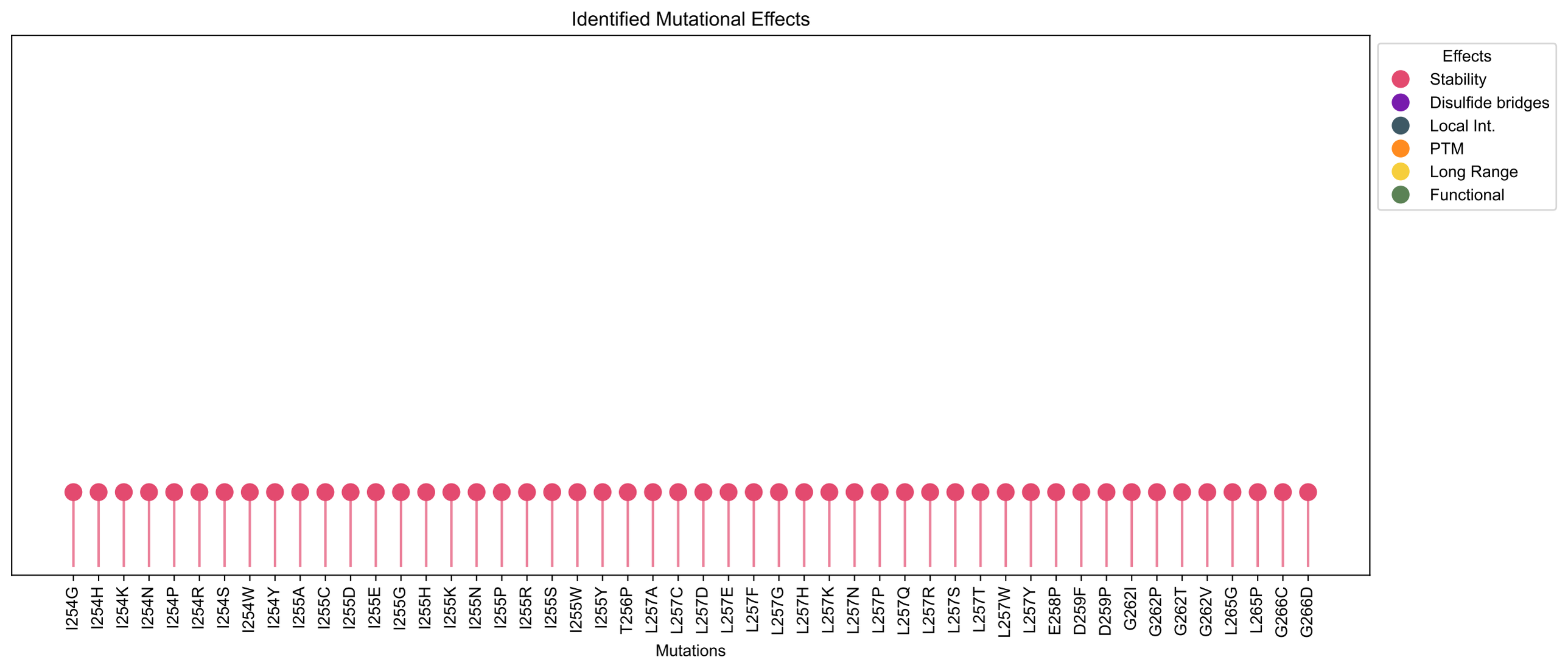

Identified Mutational Effects
Effects
Stability
Disulfide bridges
Local Int.
PTM
Long Range
Functional
S215W
S215Y
V216A
V216C
V216D
V216E
V216F
V216G
V216H
V216K
V216M
V216N
V216P
V216Q
V216R
V216S
V216T
V216W
V216Y
V217P
V218A
V218C
V218D
V218E
V218F
V218G
V218H
V218K
V218N
V218P
V218R
V218S
V218T
V218W
V218Y
Y220A
Y220C
Y220D
Y220E
Y220G
Y220H
Y220I
Y220K
Y220L
Y220M
Y220N
Y220P
Y220Q
Y220R
Y220S
Mutations

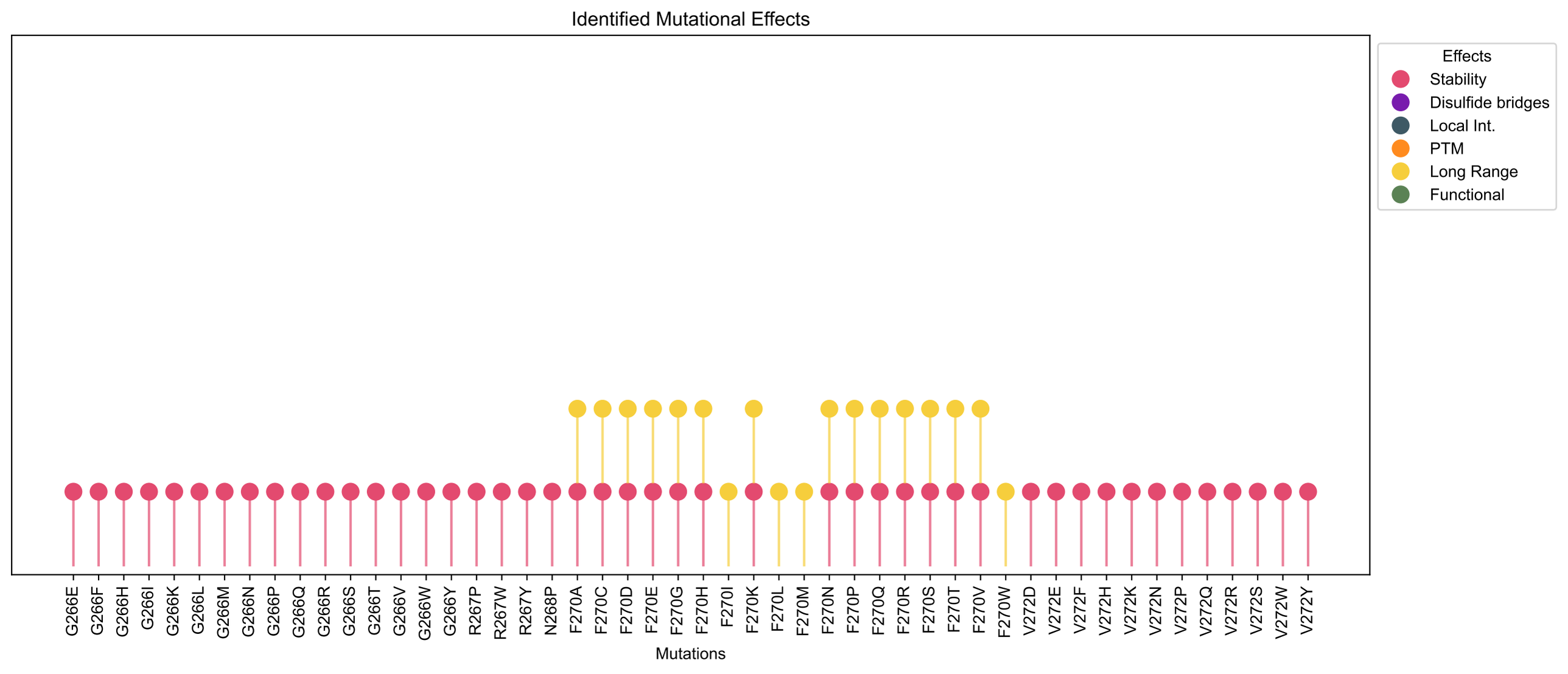

Identified Mutational Effects
Effects
Stability
Disulfide bridges
Local Int.
PTM
Long Range
Functional
Y220T
Y220V
Y220W
P222W
P223E
P223F
P223H
P223I
P223K
P223Q
P223R
P223W
P223Y
G226I
G226P
G226T
G226V
C229P
T230H
T230P
T230W
T230Y
I232A
I232C
I232D
I232E
I232G
I232H
I232K
I232N
I232P
I232Q
I232R
I232S
I232W
I232Y
H233P
Y234A
Y234C
Y234D
Y234E
Y234G
Y234H
Y234I
Y234K
Y234M
Y234N
Y234P
Y234Q
Y234R
Mutations

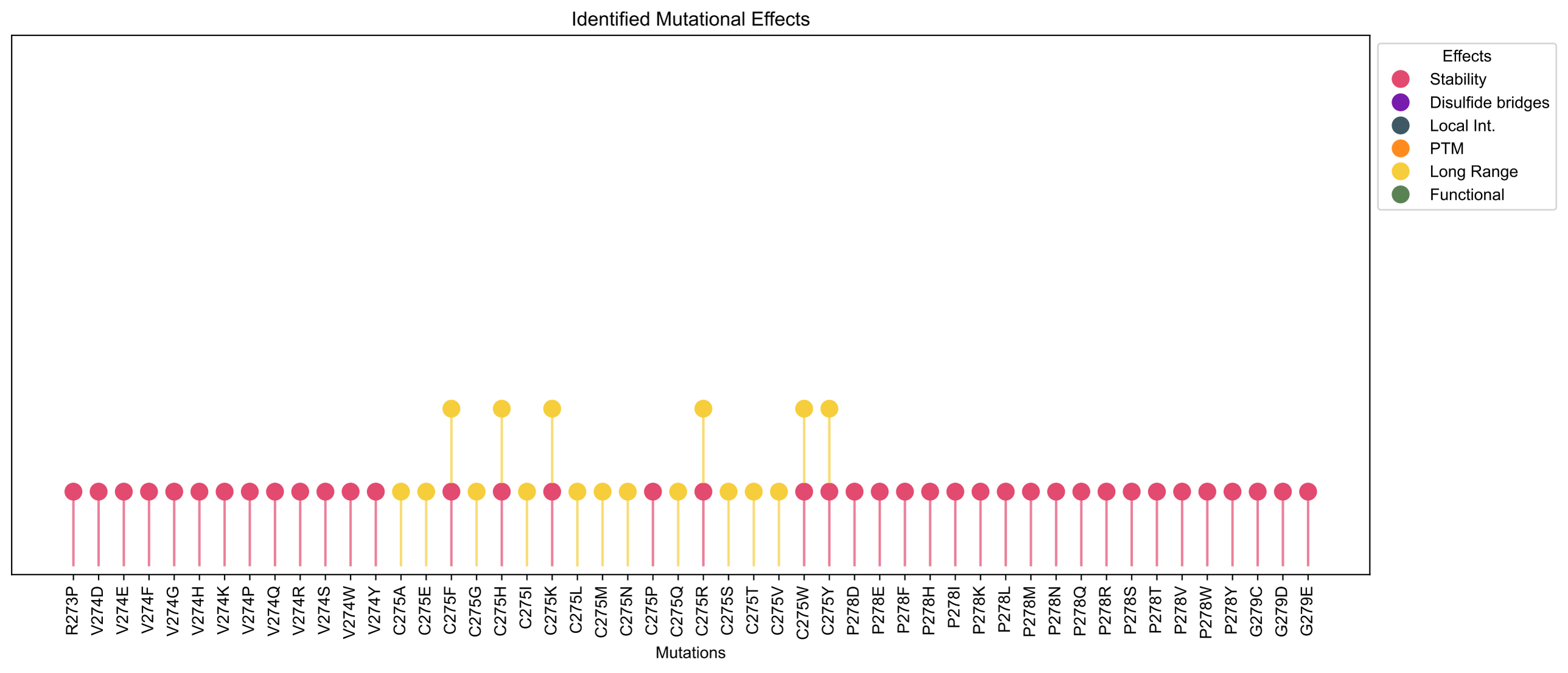

Identified Mutational Effects
Effects
Stability
Disulfide bridges
Local Int.
PTM
Long Range
Functional
Y234S
Y234T
Y234V
Y234W
N235P
Y236A
Y236C
Y236D
Y236E
Y236G
Y236H
Y236I
Y236K
Y236L
Y236M
Y236N
Y236P
Y236Q
Y236R
Y236S
Y236T
Y236V
Y236W
M237A
M237C
M237D
M237E
M237F
M237G
M237H
M237K
M237N
M237P
M237Q
M237R
M237S
M237T
M237V
M237W
M237Y
C238F
C238H
C238K
C238P
C238R
C238W
C238Y
S240H
S240P
S240W
Mutations

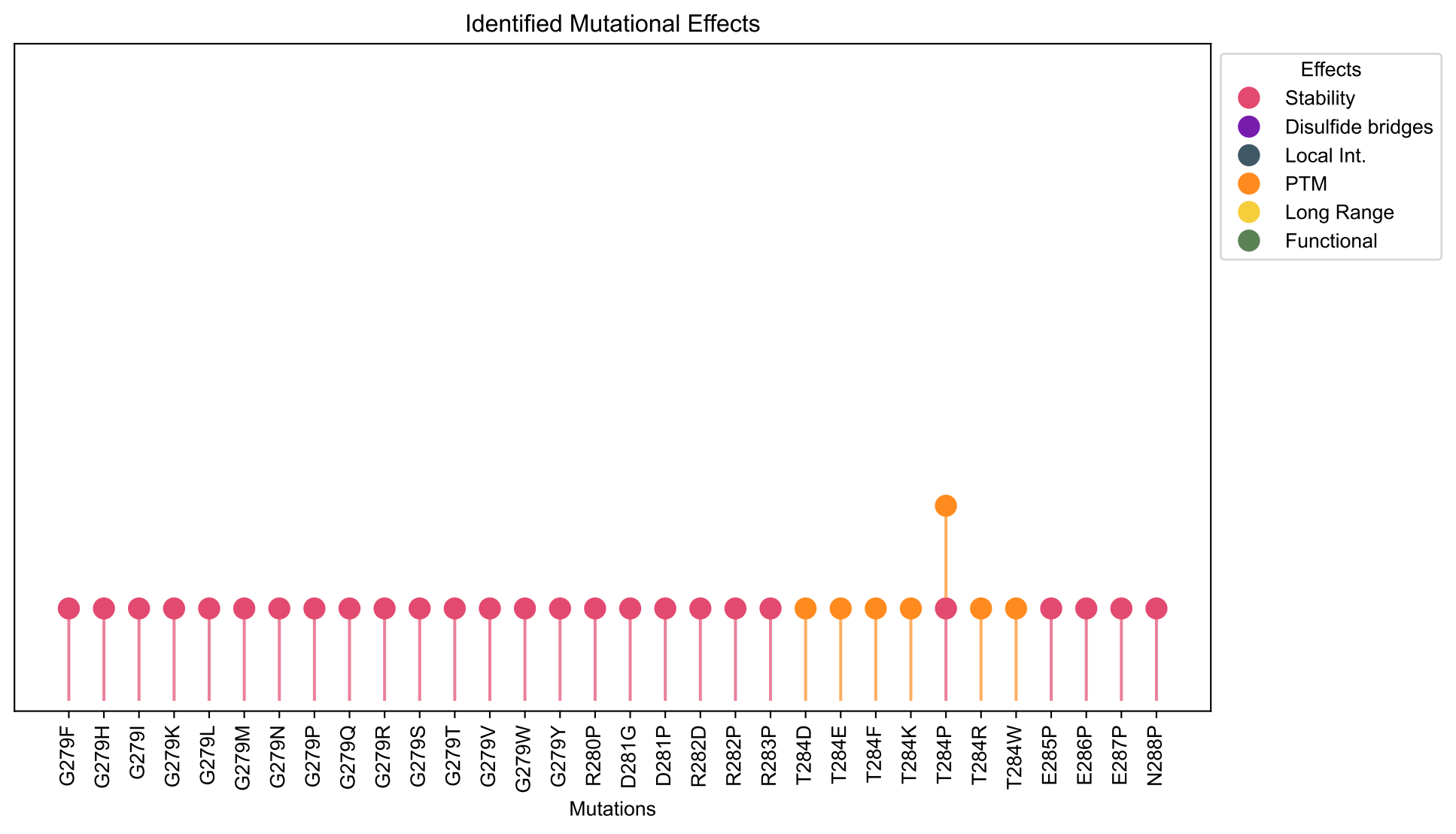

Identified Mutational Effects
Effects
Stability
Disulfide bridges
Local Int.
PTM
Long Range
Functional
S240Y
S241P
C242F
C242H
C242I
C242K
C242L
C242M
C242N
C242P
C242Q
C242R
C242T
C242V
C242W
C242Y
G244C
G244I
G244P
G244T
G244V
G245C
G245D
G245E
G245F
G245H
G245I
G245K
G245L
G245M
G245N
G245P
G245Q
G245R
G245S
G245T
G245V
G245W
G245Y
M246G
M246H
M246K
M246R
M246S
M246W
M246Y
N247I
N247P
R248P
R249A
Mutations


Identified Mutational Effects
Effects
Stability
Disulfide bridges
Local Int.
PTM
Long Range
Functional
R249C
R249D
R249E
R249F
R249G
R249H
R249I
R249K
R249L
R249M
R249N
R249P
R249Q
R249S
R249T
R249V
R249W
R249Y
I251A
I251C
I251D
I251E
I251F
I251G
I251H
I251K
I251N
I251P
I251Q
I251R
I251S
I251T
I251W
I251Y
L252D
L252E
L252G
L252H
L252P
L252S
T253G
T253K
T253P
T253R
T253W
T253Y
I254A
I254D
I254E
I254F
Mutations


Identified Mutational Effects
Effects
Stability
Disulfide bridges
Local Int.
PTM
Long Range
Functional
I254G
I254H
I254K
I254N
I254P
I254R
I254S
I254W
I254Y
I255A
I255C
I255D
I255E
I255G
I255H
I255K
I255N
I255P
I255R
I255S
I255W
I255Y
T256P
L257A
L257C
L257D
L257E
L257F
L257G
L257H
L257K
L257N
L257P
L257Q
L257R
L257S
L257T
L257W
L257Y
E258P
D259F
D259P
G262I
G262P
G262T
G262V
L265G
L265P
G266C
G266D
Mutations


Identified Mutational Effects
Effects
Stability
Disulfide bridges
Local Int.
PTM
Long Range
Functional
G266E
G266F
G266H
G266I
G266K
G266L
G266M
G266N
G266P
G266Q
G266R
G266S
G266T
G266V
G266W
G266Y
R267P
R267W
R267Y
N268P
F270A
F270C
F270D
F270E
F270G
F270H
F270I
F270K
F270L
F270M
F270N
F270P
F270Q
F270R
F270S
F270T
F270V
F270W
V272D
V272E
V272F
V272H
V272K
V272N
V272P
V272Q
V272R
V272S
V272W
V272Y
Mutations


Identified Mutational Effects
Effects
Stability
Disulfide bridges
Local Int.
PTM
Long Range
Functional
R273P
V274D
V274E
V274F
V274G
V274H
V274K
V274P
V274Q
V274R
V274S
V274W
V274Y
C275A
C275E
C275F
C275G
C275H
C275I
C275K
C275L
C275M
C275N
C275P
C275Q
C275R
C275S
C275T
C275V
C275W
C275Y
P278D
P278E
P278F
P278H
P278I
P278K
P278L
P278M
P278N
P278Q
P278R
P278S
P278T
P278V
P278W
P278Y
G279C
G279D
G279E
Mutations


Identified Mutational Effects
Effects
Stability
Disulfide bridges
Local Int.
PTM
Long Range
Functional
G279F
G279H
G279I
G279K
G279L
G279M
G279N
G279P
G279Q
G279R
G279S
G279T
G279V
G279W
G279Y
R280P
D281G
D281P
R282D
R282P
R283P
T284D
T284E
T284F
T284K
T284P
T284R
T284W
E285P
E286P
E287P
N288P
Mutations